

Steering Column Switches



Special Tool(s)	
 ST3104-A	Backprobe Pins POM 6411 or equivalent
 ST2574-A	Flex Probe Kit NU105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Ignition Switch

When the ignition lock cylinder is turned using the key, a mechanical connection positions the ignition switch to the desired position. The available ignition switch positions are:

- OFF
- ACC
- RUN
- START

When placed in any position other than OFF, the ignition switch provides fused battery voltage inputs to the Body Control Module (BCM) . When placed in the START position, the ignition switch also provides fused battery voltage input to the PCM. The ignition switch contains a key-in-ignition input that allows the BCM to detect when the key has been inserted in the ignition to initiate the module wake up cycle. This information is also used by the BCM for the brake shift interlock system. For information on the brake shift interlock system, refer to Section 307-05.

Key Removal Inhibit Solenoid

The key removal inhibit solenoid (also known as the key release interlock actuator) is an electronically controlled solenoid that prevents the ignition lock cylinder from being turned to the OFF position unless the gear selector lever is in the PARK

position. The key removal inhibit solenoid is part of the ignition switch.

Steering Column Control Module (SCCM)

The Steering Column Control Module (SCCM) controls various systems and/or features:

- Wiper and washer system, refer to Section 501-16.
- Exterior lighting, refer to Section 417-01.
- Interior lighting, refer to Section 417-02.
- Steering wheel audio controls (if equipped), refer to Refer to the appropriate section in Group 415 for the procedure..
- Steering wheel message center controls (if equipped), refer to Section 413-01.
- Cruise control, refer to Section 419-03.
- Adjustable pedals, refer to Section 206-06.
- Steering column control, refer to Section 211-04.

The SCCM also provides a pass-through circuit path for various components and/or features:

- Steering wheel switch illumination
- Driver air bag module
- Horn

The multifunction switch, brake pedal control switch (if equipped), hazard flasher switch, steering column control switch (if equipped), and LH/RH steering wheel switches (if equipped) provide inputs to the SCCM. Based on inputs from the various switches, the SCCM communicates over the High Speed Controller Area Network (HS-CAN) or the Local Interconnect Network (LIN) to control the supporting features.

Clockspring

The clockspring is mounted on the front of the SCCM and provides a continuous electrical connection between the driver air bag and the Restraints Control Module (RCM) when the steering wheel is turned. For information on the clockspring and other components of the Supplemental Restraint System (SRS), refer to Section 501-20B.

Steering Wheel Rotation Sensor

The steering wheel rotation sensor is mounted on the front of the SCCM behind the clockspring and provides steering wheel position information to the SCCM. This information is broadcast over the HS-CAN.

Multifunction Switch

The multifunction switch is mounted directly to the SCCM and controls the low/high-beam headlamps, flash-to-pass, turn signals, windshield wiper and washer and front camera washer (if equipped), by transferring the driver inputs to the SCCM. The high-beam headlamps, flash-to-pass and turn signals are communicated from the SCCM over the HS-CAN. The windshield wiper and front camera washer signals (if equipped) are communicated from the SCCM to the wiper motor internal module over the LIN. When the BCM receives the correct signal from the SCCM, the BCM activates the required component.

Brake Pedal Control Switch

The brake pedal control switch is mounted directly to the SCCM and controls the forward and rearward movement of the accelerator and brake pedals. The brake pedal control switch functions when the ignition switch is in the OFF, ACC or RUN positions. When the brake pedal control switch is pressed voltage is connected through the SCCM to the adjustable pedals motor.

Hazard Flasher Switch

The hazard flasher switch is mounted directly to the SCCM and operates when the ignition switch is in any position or when the key is removed. One press of the switch supplies a ground signal through the SCCM to the BCM. When the ground signal is sensed by the BCM it turns the directional indicators ON. A second press of the button releases the ground connection and turns the directional indicators OFF.

Steering Wheel Switches

The steering wheel switches are mounted directly to the steering wheel and wired to the SCCM . Depending on what features the vehicle is equipped with, the SCCM sends messages via the HS-CAN to the PCM (cruise control), Instrument Panel Cluster (IPC) (message center) and BCM (entertainment system). The SCCM entertainment system messages are received by the BCM via HS-CAN and gatewayed over the Medium Speed Controller Area Network (MS-CAN) where the messages are received by the applicable entertainment system module(s).

Inspection and Verification

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Multifunction switch • Ignition switch • Key efforts • Steering column shrouds	• Battery Junction Box (BJB) fuse(s): ▪ 31 (15A) ▪ 42 (5A) • Body Control Module (BCM) fuse(s): ▪ 23 (15A) ▪ 24 (15A) ▪ 28 (15A) ▪ 42 (5A) • Instrument Panel Cluster (IPC) • Steering Column Control Module (SCCM) • Passive Anti-Theft System (PATS) module • Wiring, terminals or connectors

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the ON position.
 - [GO to Pinpoint Test A](#) in this section to diagnose no power in all ignition switch positions or [GO to Pinpoint Test C](#) to diagnose no power in RUN position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00 to diagnose no response from the PCM.

7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record continuous memory DTCs.
8. Carry out the self-test diagnostics for the **BCM** and the **SCCM**.
9. If the DTCs retrieved are related to the concern, refer to the Body Control Module (BCM) DTC Chart, or the Steering Column Control Module (SCCM) DTC Chart. For all other DTCs, refer to Section 419-10.
10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B1142:29	Ignition Status 1: Signal Invalid	GO to Pinpoint Test E.
B1310:12	Run/Start Control: Circuit Short to Battery	GO to Pinpoint Test F.
B1310:14	Run/Start Control: Circuit Short to Ground or Open	GO to Pinpoint Test C.
U300A:01	Ignition Switch: General Electrical Failure	GO to Pinpoint Test A.
All other DTCs	—	REFER to the Master DTC Chart in Section 419-10.

Steering Column Control Module (SCCM) DTC Chart

DTC	Description	Action
B1007:09	High-Beam Headlamp Switch: Component Failure	REFER to Section 417-01.
B1008:09	Wiper Mode Switch: Component Failure	REFER to Section 501-16.
B1051:09	Front Washer Switch: Component Failure	REFER to Section 501-16.
B10A6:09	Main Light Switch: Component Failure	REFER to Section 417-01.
B10AD:02	Rain Sensor: General Signal Failure	REFER to Section 501-16.
B10AD:08	Rain Sensor: Bus Signal / Message Failure	REFER to Section 501-16.
B10AD:49	Rain Sensor: Internal Electronic Failure	REFER to Section 501-16.
B10AD:55	Rain Sensor: Not Configured	CARRY OUT Programmable Module Installation (PMI) Using the Integrated Diagnostic System (IDS) When the Original Module is NOT Available (using As-Built Data) on the SCCM . REFER to Section 418-01. RETRIEVE SCCM DTCs. If B10AD:55 returns, INSTALL a new SCCM . REFER to Steering Column Control Module (SCCM) in this section.

DTC	Description	Action
B1124:09	Lamp Fade Control: Component Failure	REFER to Section 417-01.
B1131:02	Wiper Motor Module: General Signal Failure	REFER to Section 501-16.
B1131:08	Wiper Motor Module: Bus Signal / Message Failure	REFER to Section 501-16.
B11D9:16	Vehicle Battery: Circuit Voltage Below Threshold	GO to Pinpoint Test J.
B11D9:17	Vehicle Battery: Circuit Voltage Above Threshold	GO to Pinpoint Test K.
B1298:09	Steering Column Adjust Up Switch: Component Failure	REFER to Section 211-04.
B1299:09	Steering Column Adjust Down Switch: Component Failure	REFER to Section 211-04.
B12A1:09	Steering Column Adjust Out Switch: Component Failure	REFER to Section 211-04.
B12A2:09	Steering Column Adjust In Switch: Component Failure	REFER to Section 211-04.
B12F7:09	Single Wipe Switch: Component Failure	REFER to Section 501-16.
B1368:11	Adjustable Pedal Position Feedback Sensor: Circuit Short to Ground	REFER to Section 206-06.
B1368:12	Adjustable Pedal Position Feedback Sensor: Circuit Short to Battery	REFER to Section 206-06.
B1369:09	Pedal Adjustment Switch: Component Failure	REFER to Section 206-06.
B1375:18	Steering Column Tilt Motor: Circuit Current Below Threshold	REFER to Section 211-04.
B1375:19	Steering Column Tilt Motor: Circuit Current Above Threshold	REFER to Section 211-04.
B1375:71	Steering Column Tilt Motor: Actuator Stuck	REFER to Section 211-04.
B137D:18	Pedal Adjustment Motor: Circuit Current Below Threshold	REFER to Section 206-06.
B137D:19	Pedal Adjustment Motor: Circuit Current Above Threshold	REFER to Section 206-06.
B137D:71	Pedal Adjustment Motor:	REFER to Section 206-06.

DTC	Description	Action
	Actuator Stuck	
B137E:18	Steering Column Telescope Motor: Circuit Current Below Threshold	REFER to Section 211-04.
B137E:19	Steering Column Telescope Motor: Circuit Current Above Threshold	REFER to Section 211-04.
B137E:71	Steering Column Telescope Motor: Actuator Stuck	REFER to Section 211-04.
B137F:09	Steering Wheel Left Switch Pack: Component Failure	REFER to Section 419-03.
B137F:11	Steering Wheel Left Switch Pack: Circuit Short To Ground	REFER to Section 419-03.
B137F:17	Steering Wheel Left Switch Pack: Circuit Voltage Above Threshold	REFER to Section 419-03.
B1380:09	Steering Wheel Right Switch Pack: Component Failure	REFER to Refer to the appropriate section in Group 415 for the procedure..
B1380:11	Steering Wheel Right Switch Pack: Circuit Short to Ground	REFER to Refer to the appropriate section in Group 415 for the procedure..
B1380:17	Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold	REFER to Refer to the appropriate section in Group 415 for the procedure..
B1C33:11	Steering Column Tilt Feedback Signal: Circuit Short To Ground	REFER to Section 211-04.
B1C33:12	Steering Column Tilt Feedback Signal: Circuit Short To Battery	REFER to Section 211-04.
B1C35:11	Steering Column Telescope Feedback Signal: Circuit Short To Ground	REFER to Section 211-04.
B1C35:12	Steering Column Telescope Feedback Signal: Circuit Short To Battery	REFER to Section 211-04.
B1D36:09	Turn Indicator Switch: Component Failure	REFER to Section 417-01.
C0051:09	Steering Wheel Position Sensor: Component Failure	INSTALL a new steering wheel rotation sensor. REFER to Section 206-09. CLEAR the DTCs. REPEAT the self-test.
C0051:16	Steering Wheel Position Sensor: Circuit Voltage	GO to Pinpoint Test I.

DTC	Description	Action
	Below Threshold	
C0051:17	Steering Wheel Position Sensor: Circuit Voltage Above Threshold	GO to Pinpoint Test I.
U1000:00	Solid State Driver Protection Active-Driver Disabled: No Sub Type Information	REPAIR all other SCCM DTCs before addressing this concern. CLEAR the DTCs and REPEAT the self test.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CARRY OUT Programmable Module Installation (PMI) Using the Integrated Diagnostic System (IDS) When the Original Module is NOT Available (using As-Built Data) on the SCCM . REFER to Section 418-01. RETRIEVE SCCM DTCs. If U2100:00 returns, INSTALL a new SCCM . REFER to Steering Column Control Module (SCCM) in this section.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	This DTC sets if the steering wheel switches do not match the switch function configuration data in the SCCM . CHECK the part numbers in Ford Catalog Advantage™ or equivalent and CONFIRM the correct SCCM and steering wheel control switches are installed to the vehicle.
U3000:49	Control Module: Internal Electronic Failure	INSTALL a new SCCM . REFER to Steering Column Control Module (SCCM) in this section. CLEAR the DTCs. REPEAT the self-test.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
No communication with the Body Control Module (BCM)	- Wiring, terminals or connectors - BCM	REFER to Section 418-00.
No power in all ignition switch positions	- Fuse - Wiring, terminals or connectors - Ignition switch - BCM	GO to Pinpoint Test A.
No power in ACC	- Fuse - Wiring, terminals or connectors - Accessory delay relay - BCM	GO to Pinpoint Test B.
No power in RUN	- Fuse - Wiring, terminals or connectors	GO to Pinpoint Test C.

	• Run/start relay • Ignition switch • BCM	
• No power in START	• Wiring, terminals or connectors • Ignition switch	• GO to Pinpoint Test D.
• The ignition key cannot be returned to the OFF position	• Wiring, terminals or connectors • Ignition lock cylinder • Key removal inhibit solenoid (part of the ignition switch) • Park range switch (part of the selector lever)	• GO to Pinpoint Test G.
• The ignition key can be turned to the OFF position when the selector lever is not in PARK	• Wiring, terminals or connectors • Key release interlock actuator (part of the ignition switch) • Park range switch (part of the selector lever)	• GO to Pinpoint Test H.
• The ignition key is hard to turn	• Ignition key • Ignition lock cylinder • Ignition switch rod • Ignition switch	• REFER to Ignition Switch — Mechanical Component Test in this section.
• The multifunction switch does not operate correctly	• Multifunction switch	• CARRY OUT the SCCM self-test. REFER to the Steering Column Control Module (SCCM) DTC Chart to diagnose any DTCs present. If no DTCs are present, INSTALL a new multifunction switch. REFER to Steering Column Multifunction Switch in this section.

• The wiper/washer switch does not operate correctly	• Wiper/washer switch	• REFER to Section 501-16.
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Pinpoint Tests

Pinpoint Test A: No Power in All Ignition Switch Positions

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/BCM for schematic and connector information.

Normal Operation

The ignition switch receives fused battery voltage from the Body Control Module (BCM) permanent voltage bus through fuse 28 (15A). In the ACC or ON position, voltage is sent to the BCM . In the START position, voltage is sent to both the BCM and PCM.

When the ignition switch is cycled to the OFF/LOCK position from any other position, the BCM begins a monitoring cycle. The monitoring cycle checks that the voltage supplied to the BCM from the ignition switch through the RUN/ACC, RUN/START and START circuits is delivered correctly for each ignition switch position.

- DTC U300A:01 (Ignition Switch: General Electric Failure) — the BCM continuously monitors the ignition switch position. This DTC sets if the voltage inputs to the BCM do not match the pre-defined configurations for each ignition switch position.

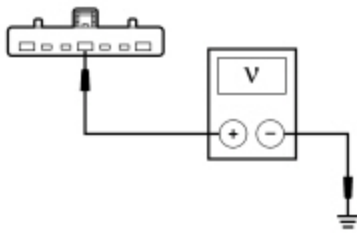
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Ignition switch

PINPOINT TEST A : NO POWER IN ALL IGNITION SWITCH POSITIONS

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.	
A1 CHECK THE IGNITION SWITCH	
• Ignition OFF. • Disconnect: Ignition Switch C250. • Carry out the Ignition Switch Component Test. Refer to Wiring Diagrams Cell 149 for component testing.	
Is the ignition switch OK?	
Yes	GO to A2 .
No	INSTALL a new ignition switch. REFER to Ignition Switch in this section. CLEAR the DTC. REPEAT the self-test. TEST the system for normal operation.
A2 CHECK FOR VOLTAGE TO THE IGNITION SWITCH	

- Measure the voltage between ignition switch [C250](#) Pin 4, circuit SBP28 (GN/RD), harness side and ground.



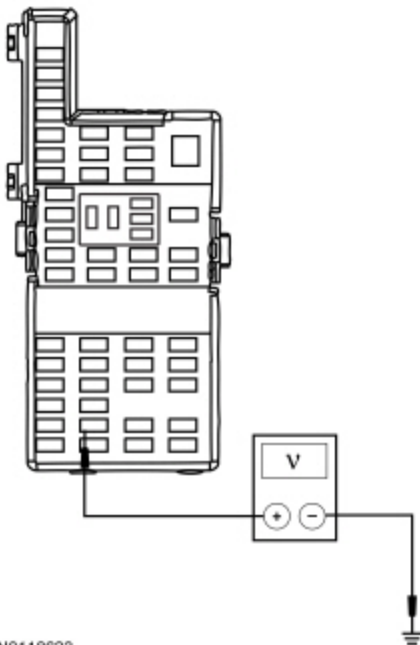
• A0089808

Is the voltage greater than 10 volts?

Yes	GO to A5 .
No	VERIFY BCM fuse 28 (15A) is OK. If OK, GO to A3 . If not OK, REFER to the Wiring Diagrams manual to identify the possible cause of the circuit short. CLEAR the DTC. REPEAT the self-test. TEST the system for normal operation.

A3 CHECK FOR VOLTAGE TO BCM

- Measure the voltage between BCM fuse 28 (15A), circuit SBP28 (GN/RD), and ground.



• N0110623

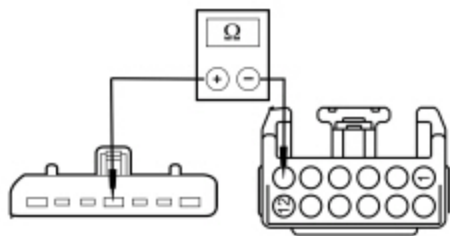
Is the voltage greater than 10 volts?

Yes	GO to A4
No	VERIFY fusible links A and B are OK. If OK, REPAIR circuit SDC04 (RD or GY). If not OK, REFER to the Wiring Diagrams manual to identify the possible cause of the circuit short. CLEAR the DTC. REPEAT the self-test. TEST the system for normal operation.

A4 CHECK FOR AN OPEN BETWEEN THE BCM AND IGNITION SWITCH

- Disconnect: BCM [C2280A](#) .

- Measure the resistance between BCM [C2280A](#) Pin 6 and ignition switch [C250](#) Pin 4, circuit SBP28 (GN/RD), harness side.



N0122571

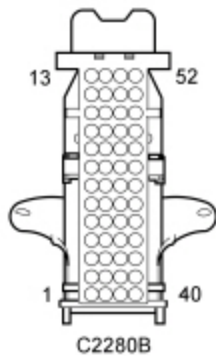
Is the resistance less than 5 ohms?

Yes	GO to A6 .
No	REPAIR circuit SBP28 (GN/RD) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

A5 CHECK THE IGNITION SWITCH OUTPUT CIRCUITS FOR AN OPEN

- Disconnect: BCM [C2280B](#) .
- Measure the resistance between ignition switch [C250](#), harness side and BCM [C2280B](#), harness side as follows:

Ignition Switch	Circuit	BCM
C250 Pin 6	CDC33 (VT/GN)	C2280B Pin 35
C250 Pin 1	CDC34 (WH/OG)	C2280B Pin 29
C250 Pin 7	CDC35 (BU/WH)	C2280B Pin 47



N0120742

Are the resistances less than 5 ohms?

Yes	GO to A6 .
No	REPAIR the affected circuit(s). CLEAR the DTC. REPEAT the self-test. TEST the system for normal operation.

A6 CHECK FOR CORRECT BCM OPERATION

- Disconnect all **BCM** connectors.
- Check the connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all **BCM** connectors and make sure they are seated correctly.
- Connect: Ignition Switch [C250](#).
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test B: No Power in ACC

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/BCM for schematic and connector information.

Normal Operation

When in ACC mode, the Body Control Module (BCM) provides ground for the coil (control) side of the run/acc relay located in the **BCM** . The run/acc relay is serviced with the **BCM** . The **BCM** also provides a ground circuit for the coil side of the accessory delay relay located in the **BCM** . The accessory delay relay can be serviced separately from the **BCM** .

This pinpoint test is intended to diagnose the following:

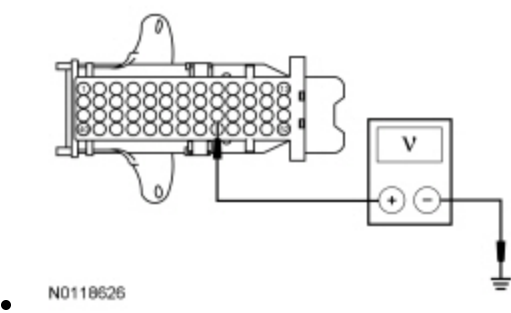
- Fuse
- Wiring, terminals or connectors
- Accessory delay relay
- **BCM**

PINPOINT TEST B : NO POWER IN ACC

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

B1 CHECK FOR ACC VOLTAGE AT THE BCM

- Disconnect: BCM [C2280B](#) .
- Ignition In ACC.
- Measure the voltage between BCM [C2280B](#) Pin 35, circuit CDC33 (VT/GN), harness side and ground.

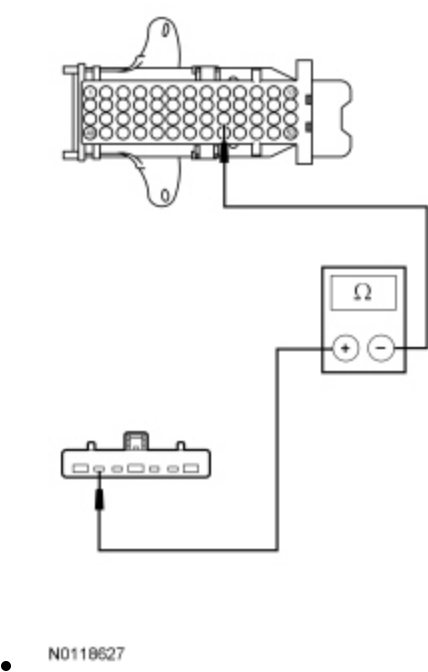


Is the voltage greater than 10 volts?

Yes	GO to B3 .
No	GO to B2 .

B2 CHECK THE IGNITION ACC CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Ignition Switch [C250](#).
- Measure the resistance between BCM [C2280B](#) Pin 35, circuit CDC33 (VT/GN), harness side and ignition switch [C250](#) Pin 6, circuit CDC33 (VT/GN), harness side.



Is the resistance less than 5 ohms?

Yes	INSTALL a new ignition switch. REFER to Ignition Switch in this section. TEST the system for normal operation.
No	REPAIR circuit CDC33 (VT/GN). TEST the system for normal operation.

B3 CHECK THE ACCESSORY DELAY RELAY

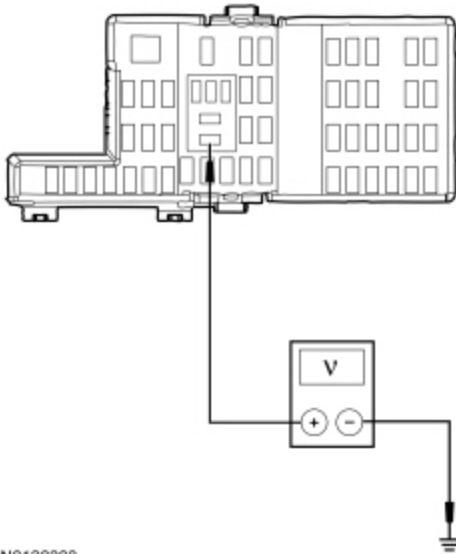
- Ignition OFF.
- Disconnect: Accessory Delay Relay .
- Carry out the appropriate relay component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Does the relay pass the test?

Yes	GO to B4 .
No	INSTALL a new accessory delay relay. TEST the system for normal operation.

B4 CHECK FOR POWER TO THE ACCESSORY DELAY RELAY

- Measure the voltage between accessory delay relay cavity 3 and ground.



N0120303

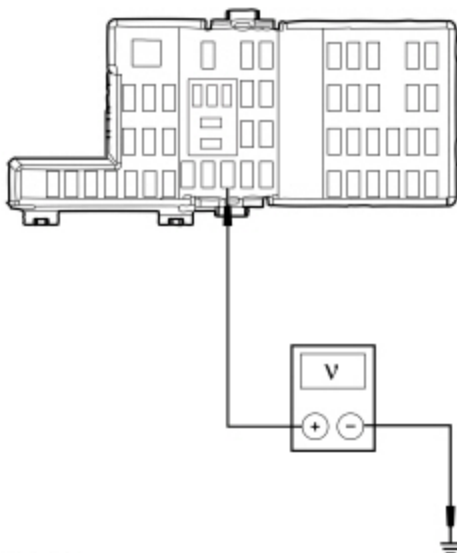
Is the voltage greater than 10 volts?

Yes	GO to B5 .
No	GO to B6 .

B5 CHECK FOR POWER TO THE RUN/ACC RELAY

- Disconnect: BCM Fuse 10 (10A) .

- Measure the voltage between BCM fuse 10 (10A), input side and ground.



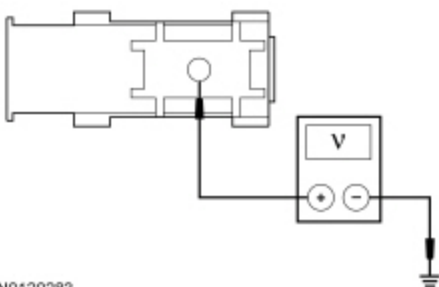
N0120282

Is the voltage greater than 10 volts?

Yes	GO to B7 .
No	GO to B6 .

B6 CHECK THE BCM BATTERY FEED CIRCUIT FOR AN OPEN

- Disconnect: BCM [C2280G](#) .
- Measure the voltage between BCM [C2280G](#) Pin 1, circuit SDC04 (RD or GY), harness side and ground.



N0120283

Is the voltage greater than 10 volts?

Yes	GO to B7 .
No	VERIFY fusible links A and B are OK. If OK, REPAIR circuit SDC04 (RD or GY). If not OK, REFER to the Wiring Diagrams manual to identify the possible cause of the circuit short. TEST the system for normal operation.

B7 CHECK FOR CORRECT BCM OPERATION

- Disconnect all BCM connectors.
- Check the connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all BCM connectors and make sure they are seated correctly.
- Connect: BCM Fuse 10 (10A) (if previously disconnected) .
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test C: No Power in RUN

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/BCM for schematic and connector information.

Normal Operation

When the ignition switch is turned to the RUN position, a voltage signal is sent to the Body Control Module (BCM) . The BCM then activates the run/start relay and the run/acc relay, which distribute fused voltage to various components. The run/acc relay is serviced as part of the BCM .

- DTC B1310:14 (Run/Start Control: Circuit Short to Ground or Open) — the BCM continuously monitors the control side of the run/start relay. This DTC sets if the BCM detects an open or a short to ground on the control circuit. The BCM uses a Field-Effect Transistor (FET) to control the run/start relay. When this DTC sets, the control side of the run/start relay is disabled via the FET . Once the condition that caused the DTC to set is repaired, the BCM must pass a self-test in order for the DTC to clear and the system to resume normal operation.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Ignition switch
- Run/start relay
- BCM

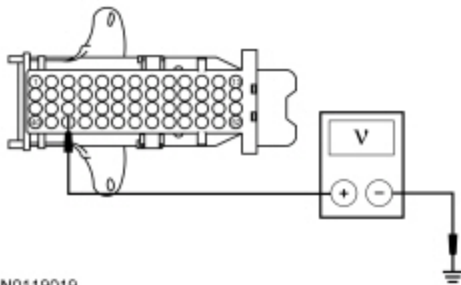
PINPOINT TEST C : NO POWER IN RUN

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.
C1 REVIEW THE BCM DTCS
• Review the BCM DTCS retrieved during inspection and verification.
Is DTC B1310:14 present?

Yes	GO to C6 .
No	For all other BCM DTCs, GO to the Body Control Module (BCM) DTC Chart. If no DTCs are present, GO to C2 .

C2 CHECK FOR RUN VOLTAGE AT THE BCM

- Ignition OFF.
- Disconnect: BCM [C2280B](#) .
- Ignition ON.
- Measure the voltage between BCM [C2280B](#) Pin 29, circuit CDC34 (WH/OG), harness side and ground.

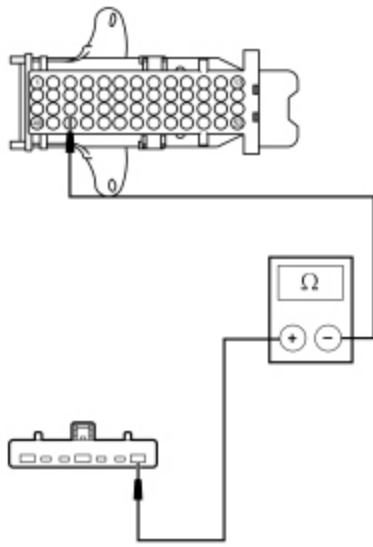


Is the voltage greater than 10 volts?

Yes	GO to C4 .
No	GO to C3 .

C3 CHECK THE IGNITION RUN CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Ignition Switch [C250](#).
- Measure the resistance between BCM [C2280B](#) Pin 29, circuit CDC34 (WH/OG), harness side and ignition switch [C250](#) Pin 1, circuit CDC34 (WH/OG), harness side.



• N0119020

Is the resistance less than 5 ohms?

Yes	INSTALL a new ignition switch. REFER to Ignition Switch in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	REPAIR open circuit CDC34 (WH/OG). TEST the system for normal operation.

C4 CHECK THE RUN/START RELAY

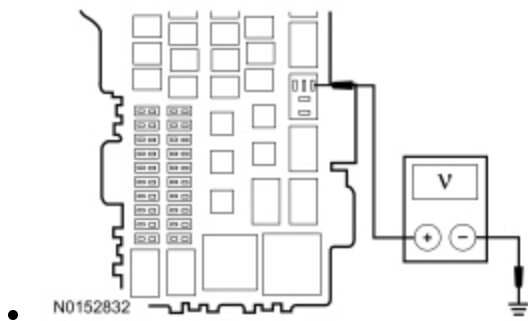
- Ignition OFF.
- Disconnect: Run/Start Relay .
- Carry out the appropriate relay component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Is the relay OK?

Yes	GO to C5 .
No	INSTALL a new run/start relay. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C5 CHECK THE VOLTAGE TO THE RUN/START RELAY COIL

- Measure the voltage between run/start relay cavity 1, circuit SBB31 (WH/RD) and ground.

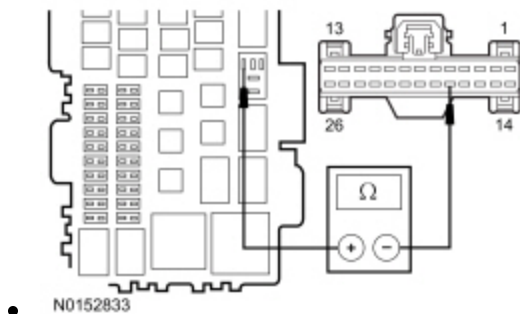


Is the voltage greater than 10 volts?

Yes	GO to C6 .
No	VERIFY BJB fuse 31 (15A) is OK. If OK, REPAIR circuit SBB31 (WH/RD). If not OK, REFER to the Wiring Diagrams manual to identify the possible cause of the circuit short. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C6 CHECK THE RUN/START RELAY CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Run/Start Relay (if not previously disconnected) .
- Disconnect: ~~BCM~~ [C2280F](#) .
- Measure the resistance between run/start relay cavity 2, circuit CDC55 (BN/VT) and ~~BCM~~ [C2280F](#) Pin 18, circuit CDC55 (BN/VT), harness side.

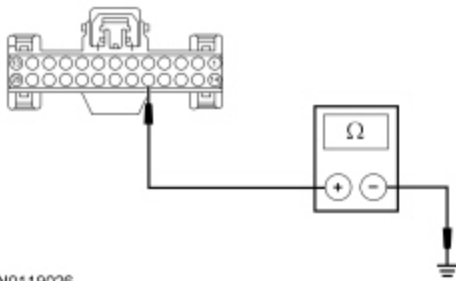


Is the resistance less than 5 ohms?

Yes	GO to C7 .
No	REPAIR circuit CDC55 (BN/VT) for an open. CARRY OUT a self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

C7 CHECK THE RUN/START RELAY CONTROL CIRCUIT FOR A SHORT TO GROUND

- Measure the resistance between ~~BCM~~ [C2280F](#) Pin 18, circuit CDC55 (BN/VT), harness side and ground.



Is the resistance greater than 10,000 ohms?

Yes	GO to C8 .
No	REPAIR circuit CDC55 (BN/VT) for short to ground. CARRY OUT a self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

C8 CHECK FOR CORRECT BCM OPERATION

- Check BCM [C2280B](#) and [C2280F](#) for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: BCM [C2280B](#) .
- Connect: BCM [C2280F](#) .
- Connect: Run/Start Relay .
- Make sure the connectors are seated correctly, then carry out a BCM self-test.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . REPEAT the self-test.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test D: No Power in START

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/BCM for schematic and connector information.

Refer to Wiring Diagrams Cell [20](#) , Starting System for schematic and connector information.

Normal Operation

When the ignition switch is turned to the START position, a voltage signal is sent to the Body Control Module (BCM) and the PCM. The BCM sends out a voltage signal to various modules to indicate a start event and the PCM checks various inputs and bus messages in an effort to start the vehicle.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Ignition switch

PINPOINT TEST D : NO POWER IN START

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

D1 CHECK THE IGNITION SWITCH

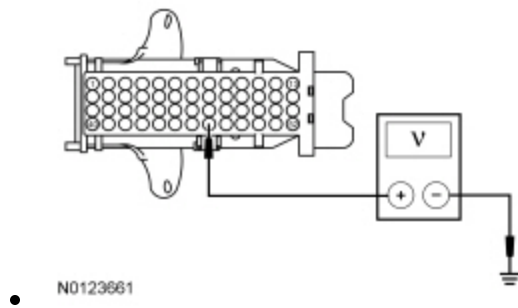
- Ignition OFF.
- Disconnect: Ignition Switch [C250](#).
- Carry out the Ignition Switch Component Test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Is the ignition switch OK?

Yes	REINSTALL the ignition switch and CONNECT the ignition switch connector removed during the component test. GO to D2 .
No	INSTALL a new ignition switch. REFER to Ignition Switch in this section. TEST the system for normal operation.

D2 CHECK THE START CIRCUIT TO THE BCM FOR VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280B](#) .
- Measure the voltage between BCM [C2280B](#) Pin 47, circuit CDC35 (BU/WH), harness side and ground, while holding the key in the START position.

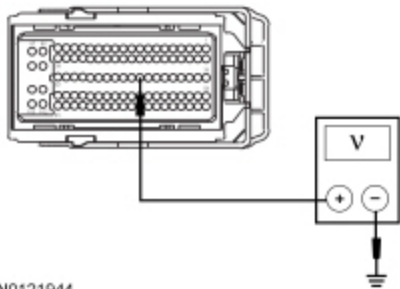


Is the voltage greater than 10 volts?

Yes	GO to D3 .
No	REPAIR circuit CDC35 (BU/WH) for an open. TEST the system for normal operation.

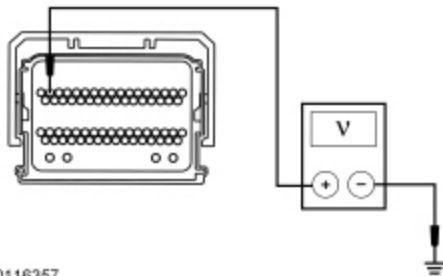
D3 CHECK THE START CIRCUIT TO THE PCM FOR VOLTAGE

- Ignition OFF.
- Disconnect: PCM [C1551B](#) (3.5L), [C175B](#) (3.7L) or [C1381B](#) (5.0L/6.2L) .
- For 3.5L, measure the voltage between PCM [C1551B](#) Pin 47, circuit CDC35 (BU/WH), harness side and ground, while holding the key in the START position.



• N0121944

- For 3.7L, 5.0L and 6.2L, measure the voltage between PCM [C175B](#) Pin 16 or [C1381B](#) Pin 16, circuit CDC35 (BU/WH), harness side and ground, while holding the key in the START position.



• N0116357

Is the voltage greater than 10 volts?

Yes	REFER to Section 303-06 to diagnose the starting system. TEST the system for normal operation.
No	REPAIR circuit CDC35 (BU/WH) for an open. TEST the system for normal operation.

Pinpoint Test E: DTC B1142:29

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/BCM for schematic and connector information.

Normal Operation

Battery voltage is sent to the ignition switch from the Battery Junction Box (BJB) fuse 28 (15A). When placed in ACC, RUN or START, the ignition switch sends fused battery voltage along 1 of 3 circuits to the Body Control Module (BCM) . Microprocessors in the BCM continuously monitor the ignition switch state.

- DTC B1142:29 (Ignition Status 1: Signal Invalid) — this DTC sets when the BCM does not receive a valid ignition state signal during the BCM self-test.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors

- Ignition switch
- BCM

PINPOINT TEST E : DTC B1142:29

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

E1 CHECK THE IGNITION SWITCH STATE (IGN_SW_STATE) PID

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — BCM .
- Monitor the IGN_SW_STATE PID while turning the ignition switch through all 4 positions (OFF, ACC, RUN and START).

Does the PID match each ignition switch position?

Yes	GO to E5 .
No	GO to E2 .

E2 CHECK THE ACC CIRCUIT FOR AN OPEN OR SHORT TO VOLTAGE

- Ignition In ACC.
- Monitor the IGN_SW_STATE PID with the ignition switch in the ACC position.

Does the PID display "Acc"?

Yes	GO to E3 .
No	If the PID displays "Off" or "Unknown", REPAIR circuit CDC33 (VT/GN) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation. If the PID displays "Run" or "Start", REPAIR circuit CDC34 (WH/OG) (RUN) or CDC35 (BU/WH) (START) for a short to voltage. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

E3 CHECK THE RUN CIRCUIT FOR AN OPEN OR SHORT TO VOLTAGE

- Ignition ON.
- Monitor the IGN_SW_STATE PID with the ignition switch in the RUN position.

Does the PID display "Run"?

Yes	GO to E4 .
No	If the PID displays "Off" or "Unknown", REPAIR circuit CDC34 (WH/OG) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation. If the PID displays "Acc" or "Start", REPAIR circuit CDC33 (VT/GN) (ACC) or CDC35 (BU/WH) (START) for a short to voltage. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

E4 CHECK THE START CIRCUIT FOR AN OPEN OR SHORT TO VOLTAGE

- Monitor the IGN_SW_STATE PID with the ignition switch in the START position.

Does the PID display "Start"?

Yes	GO to E5 .
No	If the PID displays "Off" or "Unknown", REPAIR circuit CDC35 (BU/WH) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation. If the PID displays "Acc" or "Run", REPAIR circuit CDC33 (VT/GN) (ACC) or CDC34 (WH/OG) (RUN) for a short to voltage. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

E5 CHECK FOR CORRECT BCM OPERATION

- Disconnect: All BCM Connectors .
- Check BCM connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: All BCM Connectors .
- Make sure the connectors are seated correctly, then carry out a BCM self-test.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . REPEAT the self-test. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test F: DTC B1310:12

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/SJB for schematic and connector information.

Normal Operation

When the ignition is placed in RUN mode, the Body Control Module (BCM) module provides a ground for the coil (control) side of the run/start relay located in the Battery Junction Box (BJB) . The BCM monitors the run/start relay control circuit and sets the applicable DTC if the circuit is open, shorted to ground or shorted to voltage.

- DTC B1310:12 (Run/Start Control: Circuit Short to Battery) — this DTC sets if the BCM detects a short to battery in the run/start relay control circuit during a BCM self-test or during normal operation.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Run/start relay
- BCM

PINPOINT TEST F : DTC B1310:12

F1 CHECK THE RUN/START RELAY

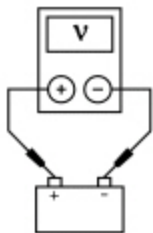
- Disconnect: Run/Start Relay .
- Carry out the appropriate relay component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Is the relay OK?

Yes	GO to F2 .
No	INSTALL a new run/start relay. CARRY OUT a self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

F2 CHECK THE RUN/START RELAY CONTROL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition ON.
- Measure and record battery voltage.



AJ0210-A

- **NOTICE:** This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool specified in the tool list in this section. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors and subsequent electrical faults

Measure the voltage by backprobing between BCM [C2280F](#) Pin 18, circuit CDC55 (BN/VT), harness side and ground.

Is the voltage equal to the recorded battery voltage?

Yes	REPAIR circuit CDC55 (BN/VT) for a short to voltage. CARRY OUT a self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.
No	GO to F3 .

F3 CHECK FOR CORRECT BCM OPERATION

- Disconnect: All BCM Connectors .

- Check the connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: All BCM Connectors .
- Connect: Run/Start Relay .
- Make sure the connectors are seated correctly, then carry out a BCM self-test.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CARRY OUT a self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CARRY OUT a self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

Pinpoint Test G: The Ignition Key Cannot Be Returned to the OFF/LOCK Position

Refer to Wiring Diagrams Cell [37](#) , Shift Interlock for schematic and connector information.

Normal Operation

The key removal inhibit solenoid (part of the ignition switch) receives battery voltage from the Body Control Module (BCM) fuse 28 (15A). When the selector lever is moved out of the PARK position, the park range switch (part of the selector lever) closes, providing a ground circuit to the key removal inhibit solenoid. The key removal inhibit solenoid activates and prevents the ignition lock cylinder from being turned to the OFF position and the key from being removed.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Key removal inhibit solenoid (part of the ignition switch)
- Ignition lock cylinder
- Brake shift interlock solenoid
- Park range switch (part of the selector lever)

PINPOINT TEST G : THE IGNITION KEY CANNOT BE RETURNED TO THE OFF/LOCK POSITION

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

NOTE: Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.

G1 CHECK FOR AN ENERGIZED KEY REMOVAL INHIBIT SOLENOID

- Remove BCM fuse 28 (15A).
- Attempt to turn the ignition lock cylinder to the RUN position and then back to the OFF/LOCK position and remove the key.

Can the ignition lock cylinder be turned to the OFF/LOCK position and the key removed?

Yes	GO to G3 .
No	GO to G2 .

G2 CHECK THE IGNITION SWITCH FOR MECHANICAL DAMAGE

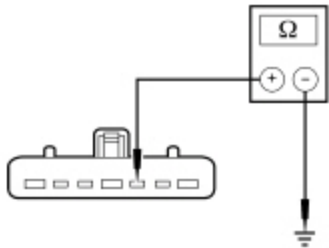
- Remove the ignition switch. Refer to Ignition Switch in this section.
- Attempt to turn the ignition lock cylinder to the OFF/LOCK position and remove the ignition key.

Does the ignition lock cylinder turn to the OFF/LOCK position and the key removed?

Yes	INSTALL a new ignition switch. REFER to Ignition Switch in this section. TEST the system for normal operation.
No	INSTALL a new ignition lock cylinder. REFER to Section 501-14. TEST the system for normal operation.

G3 CHECK THE KEY REMOVAL INHIBIT SOLENOID CIRCUIT FOR A SHORT TO GROUND

- Select PARK.
- Disconnect: Ignition Switch [C250](#).
- Measure the resistance between ignition switch [C250](#) Pin 3, circuit CDC41 (WH/BN), harness side and ground.



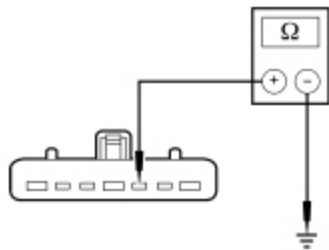
• N0065439

Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new ignition switch. REFER to Ignition Switch in this section. TEST the system for normal operation.
No	Vehicles with column shift, GO to G4 . Vehicles with floor shift, GO to G5 .

G4 ISOLATE THE BRAKE SHIFT INTERLOCK SOLENOID

- Disconnect: Brake Shift Interlock Solenoid [C2008](#).
- Measure the resistance between ignition switch [C250](#) Pin 3, circuit CDC41 (WH/BN), harness side and ground.



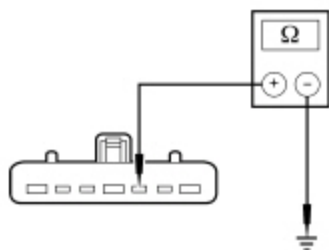
• N0065439

Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new brake shift interlock solenoid. REFER to Section 307-05. TEST the system for normal operation.
No	REPAIR circuit CDC41 (WH/BN) for a short to ground. TEST the system for normal operation.

G5 ISOLATE THE SHORT TO GROUND IN THE KEY REMOVAL INHIBIT SOLENOID CIRCUIT

- Disconnect: Floor Shifter [C3245](#).
- Measure the resistance between ignition switch [C250](#) Pin 3, circuit CDC41 (WH/BN), harness side and ground.



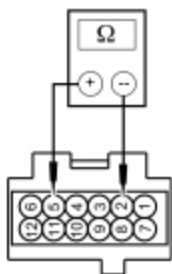
• N0065439

Is the resistance greater than 10,000 ohms?

Yes	GO to G6 .
No	REPAIR circuit CDC41 (WH/BN) for a short to ground. TEST the system for normal operation.

G6 CHECK THE PARK DETECT SWITCH

- Measure the resistance between the selector lever C3245-2, component side and selector lever C3245-5, component side with and without the selector lever in the PARK range and the selector lever button released.



• N0151254

Is the resistance greater than 10,000 ohms with the selector lever in PARK and less than 5 ohms with the selector lever in REVERSE, NEUTRAL and DRIVE?

Yes	CHECK the selector lever cable adjustment. REFER to Section 307-05. INSPECT the selector lever, selector lever knob, and selector lever cable for damage. INSTALL new components as necessary. REFER to Section 307-05. TEST the system for normal operation.
No	INSTALL a new selector lever. REFER to Section 307-05. TEST the system for normal operation.

Pinpoint Test H: The Ignition Key Can Be Turned to the OFF/LOCK Position When the Selector Lever is Not In PARK

Refer to Wiring Diagrams Cell 37 , Shift Interlock for schematic and connector information.

Normal Operation

The key removal inhibit solenoid (part of the ignition switch) receives battery voltage from Body Control Module (BCM) fuse 28 (15A). When the selector lever is moved out of the PARK position, the park range switch (part of the selector lever) closes, providing a ground circuit to the key removal inhibit solenoid. The key removal inhibit solenoid activates and prevents the ignition lock cylinder from being turned to the OFF/LOCK position and the key from being removed.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Key release interlock actuator (part of the ignition switch)
- Park range switch (part of the selector lever)

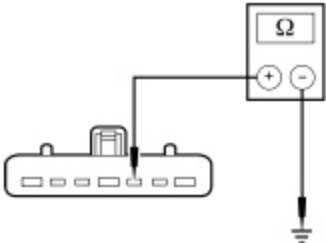
PINPOINT TEST H : THE IGNITION KEY CAN BE TURNED TO THE OFF/LOCK POSITION WHEN THE SELECTOR LEVER IS NOT IN PARK

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

NOTE: Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.

H1 CHECK THE KEY INHIBIT SOLENOID CIRCUIT FOR GROUND

- Disconnect: Ignition Switch C250.
- Select DRIVE.
- Measure the resistance between ignition switch C250 Pin 3, circuit CDC41 (WH/BN), harness side and ground.



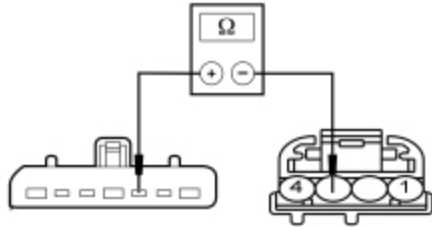
N0065439

Is the resistance less than 5 ohms?

Yes	INSTALL a new ignition switch. REFER to Ignition Switch in this section. TEST the system for normal operation.
No	Vehicles with column shift, GO to H2 . Vehicles with floor shift, GO to H3 .

H2 CHECK THE KEY INHIBIT SOLENOID CIRCUIT FOR AN OPEN (COLUMN SHIFT)

- Disconnect: Brake Shift Interlock Solenoid [C2008](#).
- Measure the resistance between ignition switch [C250](#) Pin 3, circuit CDC41 (WH/BN), harness side and brake shift interlock solenoid [C2008](#) Pin 3, circuit CDC41 (WH/BN), harness side.



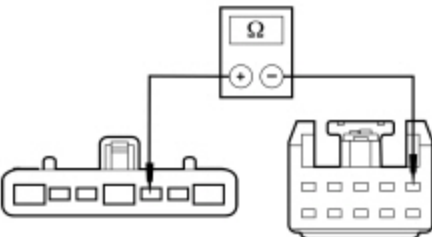
N0122615

Is the resistance less than 5 ohms?

Yes	INSTALL a new brake shift interlock solenoid. REFER to Section 307-05. TEST the system for normal operation.
No	REPAIR circuit CDC41 (WH/BN) for an open. TEST the system for normal operation.

H3 CHECK THE KEY INHIBIT SOLENOID CIRCUIT FOR AN OPEN (FLOOR SHIFT)

- Disconnect: Floor Shifter [C3245](#).
- Measure the resistance between ignition switch [C250](#) Pin 3, circuit CDC41 (WH/BN), harness side and floor shifter [C3245](#) Pin 1, circuit CDC41 (WH/BN), harness side.



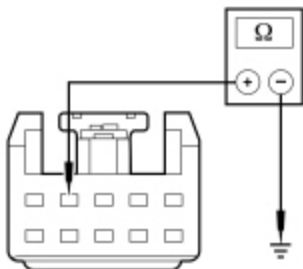
N0122061

Is the resistance less than 5 ohms?

Yes	GO to H4 .
No	REPAIR circuit CDC41 (WH/BN) for an open. TEST the system for normal operation.

H4 CHECK THE FLOOR SHIFTER GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between floor shifter [C3245](#) Pin 5, circuit GD138 (BK/WH), harness side and ground.



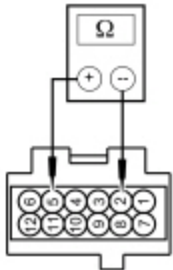
N0119174

Is the resistance less than 5 ohms?

Yes	GO to H5 .
No	REPAIR circuit GD138 (BK/WH) for an open. TEST the system for normal operation.

H5 CHECK THE PARK DETECT SWITCH

- Measure the resistance between the selector lever C3245-2, component side and selector lever C3245-5, component side with and without the selector lever in the PARK range and the selector lever button released.



N0151254

Is the resistance greater than 10,000 ohms with the selector lever in PARK and less than 5 ohms with the selector lever in REVERSE, NEUTRAL and DRIVE?

Yes	CHECK the selector lever cable adjustment. REFER to Section 307-05. INSPECT the selector lever, selector lever knob, and selector lever cable for damage. INSTALL new components as necessary. REFER to Section 307-05. TEST the system for normal operation.
No	INSTALL a new selector lever. REFER to Section 307-05. TEST the system for normal operation.

Pinpoint Test I: DTCs C0051:16 And C0051:17

Normal Operation

The Steering Column Control Module (SCCM) continuously monitors the voltage supplied to the steering wheel rotation sensor (also known as the steering wheel position sensor). If voltage outside of defined limits is detected by the SCCM , the

applicable DTC sets.

- DTC C0051:16 (Steering Wheel Position Sensor: Circuit Voltage Below Threshold) — If the SCCM detects voltage to the steering wheel rotation sensor below 10 volts this DTC sets.
- DTC C0051:17 (Steering Wheel Position Sensor: Circuit Voltage Above Threshold) — If the SCCM detects voltage to the steering wheel rotation sensor above 15.5 volts this DTC sets.

This pinpoint test is intended to diagnose the following:

- Steering wheel rotation sensor

PINPOINT TEST I : DTCS C0051:16 AND C0051:17

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

I1 CHECK THE SCCM DTCS

- Review the SCCM DTCs retrieved and recorded during Inspection and Verification.

Is DTC B119D:16 and/or B119D:17 present?

Yes	DIAGNOSE DTCs B119D:16 and B119D:17 before diagnosing DTCs C0051:16 or C0051:17. For DTC B119D:16, GO to Pinpoint Test J . For DTC B119D:17, GO to Pinpoint Test K .
No	GO to I2 .

I2 CHECK THE STEERING WHEEL ANGLE SENSOR STATUS (SW_ANG_SENS) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — SCCM .
- Monitor the SW_ANG_SENS PID.

Does the PID display CORRECT?

Yes	The condition that caused the DTC to set is not present at this time. CLEAR the DTCs. DRIVE the vehicle for more than 1 minute at a speed greater than 32 km/h (20 mph) and make at least one left turn and one right turn. REPEAT the self-test. If the DTC(s) does not return, RETURN the vehicle to the customer. If the DTC(s) return, GO to I3 .
No	If the PID displays FAULT, INSTALL a new steering wheel rotation sensor. REFER to Section 206-09. REPEAT the self-test. TEST the system for normal operation. If the PID displays PSCM NOT INITIALIZED, GO to I3 .

I3 CHECK FOR CORRECT SCCM OPERATION

- Ignition OFF.
- Disconnect: All SCCM Connectors .
- Check the SCCM connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: All SCCM Connectors .

- Make sure the **SCCM** connectors are seated correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Steering Column Control Module (SCCM) in this section. CLEAR the DTCs. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test J: DTC B119D:16

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/SJB for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) continuously monitors input voltage for correct operation. If voltage outside of defined limits is detected by the **SCCM** , the applicable DTC sets.

- DTC B119D:16 (Vehicle Battery: Circuit Below Threshold) — This DTC sets in continuous memory if the **SCCM** detects battery voltage below 8 volts on the battery voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Fuses
- Wiring, terminals or connectors
- Charging system
- Vehicle battery
- **SCCM**

NOTE: Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.

NOTE: DTC B119D:16 can set if the vehicle battery has been discharged. The vehicle battery may become discharged due to excessive load(s) on the charging system from aftermarket accessories or if the battery has been left unattended with the accessories on.

PINPOINT TEST J : DTC B119D:16

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

J1 VERIFY THE SCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **SCCM** .
- Clear the DTCs and repeat the self-test.

Is DTC B119D:16 still present?

Yes	GO to J2 .
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No	The system is operating normally at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
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J2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM

- Enter the following diagnostic mode on the scan tool: Self Test — PCM .
- Retrieve and record all continuous memory DTCs from the PCM.

Are any charging system DTCs present in the PCM?

Yes	REFER to Section 414-00 to diagnose the charging system DTCs.
No	GO to J3 .

J3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE

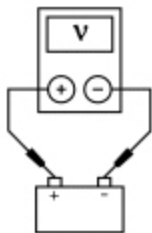
- Ignition OFF.
- Refer to Section 414-01 and carry out pinpoint test A: Battery Condition Test.

Is the battery OK and fully charged?

Yes	GO to J4 .
No	REFER to Section 414-00 to diagnose the charging system.

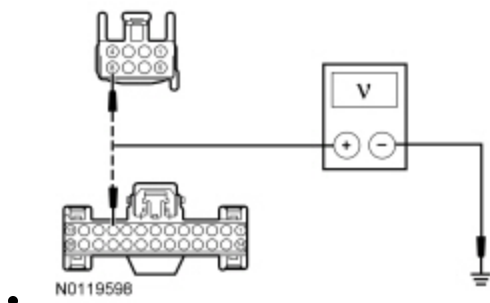
J4 CHECK THE SCCM VOLTAGE SUPPLY

- Measure and record battery voltage.



AJ0210-A

- Disconnect: ~~SCCM~~ [C2414A](#) .
- Disconnect: ~~SCCM~~ [C2414B](#) .
- Measure the voltage between ground and:
 - ~~SCCM~~ [C2414A](#) Pin 10, SBP23 (WH/RD), harness side.
 - ~~SCCM~~ [C2414B](#) Pin 8, SBP24 (VT/RD), harness side.

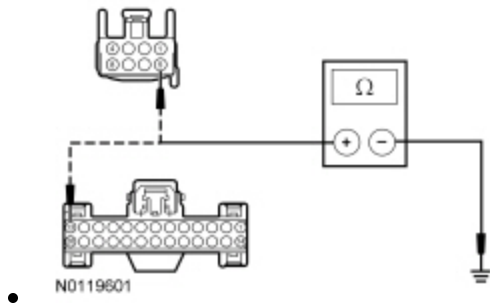


Are the voltages within 0.2 volt of the recorded battery voltage?

Yes	GO to J5 .
No	VERIFY BCM fuses 23 (15A) and 24 (15A) are OK. If OK, REPAIR the affected circuit(s). If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

J5 CHECK THE SCCM GROUND CIRCUIT

- Disconnect the battery negative cable. Refer to Section 414-01.
- Measure the resistance between ground and:
 - SCCM [C2414A](#) Pin 13, GD133 (BK), harness side.
 - SCCM [C2414B](#) Pin 5, GD138 (BK/WH), harness side.



Are the resistances less than 5 ohms?

Yes	CONNECT the battery negative cable. GO to J6 .
No	REPAIR the affected circuit(s). CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

J6 CHECK FOR CORRECT SCCM OPERATION

- Disconnect: All SCCM Connectors .
- Check the SCCM connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: All SCCM Connectors .

- Make sure the SCCM connectors are seated correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Steering Column Control Module (SCCM) in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test K: DTC B119D:17

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/SJB for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) continuously monitors input voltage for correct operation. If voltage outside of defined limits is detected by the SCCM , the applicable DTC sets.

- DTC B119D:17 (Vehicle Battery: Circuit Above Threshold) — This DTC sets in continuous memory if the SCCM detects battery voltage above 19 volts on the battery voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Charging system
- SCCM

NOTE: Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.

NOTE: DTC B119D:17 can set if the vehicle has been recently jump started or the vehicle battery has been recently charged.

PINPOINT TEST K : DTC B119D:17

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

K1 CHECK FOR HIGH BATTERY VOLTAGE AND/OR CHARGING SYSTEM DTC IN OTHER MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .
- Retrieve and recorded all CMDTCs from all modules.

Are any high battery voltage and/or charging system DTCs present in other modules?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to K2 .

K2 CHECK THE BATTERY VOLTAGE

- Turn off all interior/exterior lights and accessories.
- Start and run the engine at approximately 2,000 rpm for 3 minutes while monitoring the battery voltage.

Does the battery voltage rise to 16.5 volts or higher?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to K3 .

K3 RECHECK FOR SCCM DTC B119D:17

- Turn the engine off.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — SCCM .
- Clear the continuous memory DTCs and repeat the self-test.

Is DTC B119D:17 present in the SCCM ?

Yes	GO to K4 .
No	The system is operating normally at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

K4 CHECK FOR CORRECT SCCM OPERATION

- Ignition OFF.
- Disconnect: All SCCM Connectors .
- Check the SCCM connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: All SCCM Connectors .
- Make sure the SCCM connectors are seated correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new <u>SCCM</u> . REFER to Steering Column Control Module (SCCM) in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal

Component Tests

Ignition Switch — Mechanical

The following conditions can cause difficulty in operating the ignition switch and lock cylinder:

- Burrs on the lock cylinder key
- Insufficient lube on the lock cylinder
- Binding lock cylinder
- Burrs or foreign material around the rack and pinion actuator in the lock cylinder housing
- Insufficient lube on the actuator (Do not apply lubricant to the inside of the ignition switch.)
- Binding ignition switch

If the steering wheel lock is engaged with the wheels loaded against a curb, high effort is necessary to turn the key from lock. Turn the steering wheel to either side of the lock to unload the system.

Carry out the following test to determine if the ignition switch and lock cylinder are operating correctly.

1. Inspect the ignition key for any burrs, damage or incorrect cut. Replace the ignition key as necessary.
2. **NOTE:** *The steering wheel may be locked full left or full right. If the steering wheel is locked, it is necessary to apply turning effort to the steering wheel in the direction of the lock while turning the key.*

Turn the key to the ACC position and then the RUN position.

- If the ignition key turns to the ACC and RUN position, continue with Step 3.
- If the ignition key does not turn to the ACC and RUN position, continue with Step 4.

3. **NOTE:** *The ignition switch and lock cylinder should return from the START position back to the RUN position without assistance.*

Turn the ignition key to the START position and release the key.

- If the ignition switch and lock cylinder return from the START position back to the RUN position without assistance, the ignition switch is operating correctly at this time.
- If the ignition switch and lock cylinder do not return from the START position back to the RUN position without assistance, continue with Step 4.

4. Remove the ignition lock cylinder. Refer to Section 501-14.
5. Rotate the ignition lock cylinder through all of the switch positions.
 - If the lock cylinder operates correctly, continue with Step 6.
 - If the lock cylinder does not operate correctly, install a new ignition lock cylinder. Refer to Section 501-14.
6. Check for binding or sticking ignition switch actuating rod, burrs around the rack-and-pinion actuator in the ignition lock cylinder housing or insufficient lubrication.
 - If there is sufficient lubrication and there are no burrs, binding or sticking conditions, install a new ignition switch. Refer to Ignition Switch in this section.
 - If there is insufficient lubrication, burrs, binding or sticking conditions, repair or lubricate as necessary.